Performance Evaluation of SVM

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1 SVM Project: Project and Dataset Description

This project focuses on the application of a Support Vector Machine (SVM) to solve a multiclass classification problem, with the objective of predicting professional experience level. The dataset "dataset_salary_2024.csv" contains salary information for employees from multiple countries, including attributes such as work year, job title, company size, remote work ratio, salary in USD, employee residence, and experience level. These characteristics were grouped into 3 categories:

1.1 Continuous Variables

– **work_year** — chronological indicator that reflects economic trends which influence salary policies, indirectly determining experience level
– **salary_in_usd** — a real numerical value highly relevant to the model, as salary tends to increase with the project's target variable, *experience_level*.

1.2 Categorical Variables

The categorical variables — *job_title*, *residence_grouped*, *remote_ratio*, and *company_size* — were transformed using One-Hot Encoding, as the classification model cannot directly process textual values. This encoding converts each category into a binary column (True/False).

The analysis of country distributions revealed strongly underrepresented classes. To reduce noise and prevent negative effects on model training, countries with fewer than 50 employees were aggregated into a single category "Other", resulting in a new preprocessed column *residence_grouped*.

1.3 Removal of Redundant Attributes

Attributes such as *salary*, *salary_currency*, *employee_residence*, *company_location*, and *job_group* were removed because: *salary* and *salary_currency* duplicate the information contained in *salary_in_usd*; *employee_residence* was only used to construct the preprocessed *residence_grouped*; *company_location* mirrors *employee_residence*; *job_group* ends not being relevant, as *job_titles* can not be grouped due to lack of name homogens.

1.4 Plot Analysis

The preliminary 'plot 3' shows a proportional increase in *salary_in_usd* across experience levels, confirming that salary serves as an indirect indicator of seniority. Plotting the graphs in relation to *salary_in_usd* helps identify the usefulness of the other variables and detect anomalies in categories that do not influence the experience level.

Plotting *salary_in_usd* against *job_title* ('plot 7–13') reveals that keywords such as "Lead" or "Manager" may refer to roles with different levels of seniority, generating an inconsistent signal. Due to high lexical variability and mixed semantics, job titles cannot be reliably grouped for classification purposes.

Plotting *salary_in_usd* against *work_year* (plot 4), *employee_residence* (plot 5), and *company_size* (plot 6) shows that salaries generally increase over time (recent years → higher pay), employees in developed countries tend to earn more, and larger companies typically offer higher compensation.

Median salary in USD across experience levels

1.5 SVM Modeling and Hyperparameter Optimization

This step involves constructing a search grid containing possible values for the SVM hyperparameters. The regularization coefficient C was assigned three levels of flexibility (0.5 - moderate, 1 - standard, 5 - low regularization), and two kernel types: linear and RBF. The γ parameter shows the influence a single training sample has on the decision function.

A *GridSearchCV* object was created to evaluate all hyperparameter combinations using 3-fold cross-validation and compute each model's performance. The parameter *n_jobs = -1* enables the use of all CPU cores, allowing multiple combinations to be tested in parallel ($3 \times 2 \times 2 = 12$ hyperparameter combinations, each evaluated in 3 folds → 36 trained models).

Best params: C=5, gamma='scale', kernel='rbf'

The combination of $C = 5$, $\gamma =$ 'scale', and the RBF kernel indicates that the SVM model uses a nonlinear, stricter, and more stable decision boundary to separate the experience-level classes.

With the RBF kernel, the decision function becomes:

$$f(x) = \sum_{i=1}^{N} \alpha_i y_i K(x_i, x) + b$$

In the constructed model, the optimal choice is the RBF kernel because: The interaction between the variables and the target is not linear; Seniority appears to increase linearly with *salary_in_usd*, but the plots reveal only a general trend rather than a perfectly linear relationship; *One-Hot Encoding* produces a high-dimensional feature space where the experience-level classes cannot be separated using linear methods.

1.6 Training the Data

During the training and testing stage, the dataset was divided into independent variables (X) and the target variable (y), corresponding to *experience_level*. The data split was performed using an 80% training set and a 20% test set, applying the *train_test_split* method with stratification. The parameter *stratify* $= y$ was used to ensure that the experience level classes (EN, MI, SE, EX) were preserved in the same proportions.

1.7 Results Analysis for Multiple Experimental Scenarios

Initially, it was hypothesized that the *job_title* column, when processed using One-Hot Encoding, could significantly increase dimensional complexity due to its high lexical variability. This would lead to increased computational cost and a higher risk of overfitting.

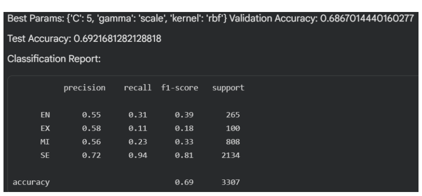
Exploratory analysis revealed a total of 155 distinct job titles, many of which appear rarely, introducing noise into the classification process. For this reason, three experimental scenarios were evaluated.

Complete Removal of the *job_title* Column

The complete removal of the *job_title* column resulted in an accuracy of 67%, indicating that *job_title* plays an important role in experience-level classification.

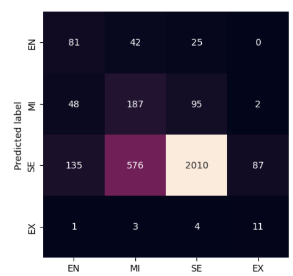
Preprocessing *job_title* into a *job_group* Column

Preprocessing *job_title* into a new *job_group* feature resulted in an accuracy of 69%. The confusion matrix shows that the model learns the dominant SE class significantly better (2010 correct classifications). Many MI employees are predicted as SE (576), EN is confused with MI (48) and SE (135), and EX is absorbed by SE (87). EN and EX show weaker performance due to underrepresentation and class imbalance.


Best Params: {'C': 5, 'gamma': 'scale', 'kernel': 'rbf'} Validation Accuracy: 0.6867014440160277
Test Accuracy: 0.6921681282128818
Classification Report:

	precision	recall	f1-score	support
EN	0.55	0.31	0.39	265
EX	0.58	0.11	0.18	100
MI	0.56	0.23	0.33	808
SE	0.72	0.94	0.81	2134
accuracy			0.69	3307

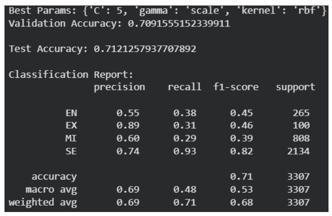
Results with job group



Confusion Matrix Job Group

Retaining the Original *job_title* Column

The best classification performance was achieved when the *job_title* column was kept unchanged, resulting in an accuracy of 71%. Although SE correct classifications slightly decrease (from 2010 to 1987), performance improves for EN (100 correct predictions out of 265) and EX (31 correct predictions out of 100). The MI class continues to be confused with SE due to similar determining features, while the SE class remains dominant and absorbs underrepresented classes.



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Best Params: {'C': 5, 'gamma': 'scale', 'kernel': 'rbf'}
Validation Accuracy: 0.7091555152339911

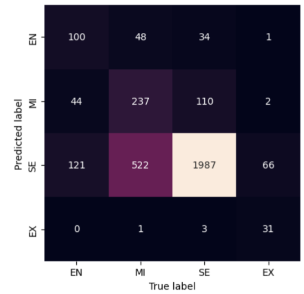
Test Accuracy: 0.7121257937707892

Classification Report:
              precision    recall  f1-score   support

          EN       0.55      0.38      0.45       265
          EX       0.89      0.31      0.46       100
          MI       0.60      0.29      0.39       808
          SE       0.74      0.93      0.82      2134

    accuracy                           0.71      3307
   macro avg       0.69      0.48      0.53      3307
weighted avg       0.69      0.71      0.68      3307
```

Job Title Results



Confusion Matrix Job Title

2 Bibliography

- Course materials Regression, Classification, Unsupervised Learning by Prof. Nicola Conci
- http://scikit-learn.org/
- https://www.geeksforgeeks.org/
- https://www.mygreatlearning.com/blog/introduction-to-support-vector-machine/